

Question Paper

Exam Date & Time: 12-Nov-2019 (10:00 AM - 01:00 PM)



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

1st Semester of B.Sc.(Physics) Examination November-2019

Date: 12/11/2019 Time: 10.00 a.m to 1.00 p.m

MATHEMATICAL PHYSICS-I [PD101]

Marks: 70

Duration: 180 mins.

Multiple Choice Questions

Answer all the questions.

Section Duration: 30 mins

Q-I Choose the correct answer for the following questions.

- 1) A function $f(x) = |x|$ is _____. (1)

☐ differentiable only at origin
 ☐ not differentiable at origin
 ☐ continuous only at origin
 ☐ not continuous at origin

- 2) The order and degree of the differential equation $1 + \left(\frac{dy}{dx}\right)^3 = \left(\frac{d^4y}{dx^4}\right)^7$ are _____ respectively. (1)

☐ 4 and 7
 ☐ 7 and 4
 ☐ 1 and 3
 ☐ 3 and 1

- 3) Which of the following differential equation has solution $y = kx + k^2$, where k is the arbitrary constant? (1)

☐ $y = \frac{dy}{dx} + \left(\frac{dy}{dx}\right)^2$
☐ $y = x \frac{dy}{dx} + \left(\frac{dy}{dx}\right)^2$
☐ $y = \left(\frac{dy}{dx}\right)^2$
☐ $y = \left(\frac{dy}{dx}\right)^2 + x$

- 4) If $2 + 3i$ and $2 - 3i$ are roots of the auxiliary equation of the homogenous differential equation, then general solution of this differential equation is _____, where c_1 and c_2 are arbitrary constants. (1)

☐ $c_1 e^{2x} + c_2 e^{-3x}$
☐ $c_1 e^{2x} + c_2 e^{3x}$
☐ $(c_1 \cos 3x + c_2 \sin 3x) e^{2x}$
☐ $(c_1 \cos 2x + c_2 \sin 2x) e^{3x}$

- 5) The Wronskian of two functions $\cos 3x$ and $\sin 3x$ is _____. (1)

☐ 0
 ☐ 6
 ☐ 9
 ☐ 3

- 6) $\lim_{n \rightarrow \infty} \frac{3n^3 + 2n^2 - 8}{2n^4 + n - 7} =$ _____. (1)

☐ 2
 ☐ 0
 ☐ $\frac{8}{7}$
☐ Not exist

- 7) If $\vec{a}, \vec{b}$ and $\vec{c}$ are three co-terminous edges of parallelepiped, then its volume is _____. (1)

☐ $\vec{a} \cdot (\vec{b} \times \vec{c})$
☐ $\vec{a} \cdot (\vec{b} \cdot \vec{c})$
☐ $\vec{a} \times (\vec{b} \times \vec{c})$
☐ $(\vec{a} \times \vec{b}) \times \vec{c}$

- 8) Two vectors $\vec{a}$ and $\vec{b}$ are perpendicular, if _____. (1)

☐ $\vec{a} \cdot \vec{b} = -1$
☐ $\vec{a} \cdot \vec{b} = 0$
☐ $\vec{a} \times \vec{b} = -1$
☐ $\vec{a} \times \vec{b} = 0$

- 9) A maximum value of the directional derivative of scalar point function $f(x, y, z)$ is _____. (1)

- 10) If $\vec{A}$ is such that $\nabla \times \vec{A} = \vec{0}$, then $\vec{A}$ is called _____. (1)
- 11) If $\vec{F}$ is the velocity of a fluid particle, then $\int_C \vec{F} \cdot d\vec{r}$ represents _____ of the fluid. (1)
- 12) Which of the following result established relation between surface integral and volume integral? (1)
- 13) _____ is normal to the surface $f(x, y, z)$. (1)
- 14) What is the probability that number selected at random from the numbers (1,2,3,...,15) is a multiple of 4? (1)
- 15) Conditional probability of two independent events A and B can be written as _____. (1)
- 16) Which of the following is a continuous probability distribution? (1)
- 17) For cylindrical coordinates $x = r \cos \theta, y = r \sin \theta, z = z$, what is r ? (1)
- 18) A hemisphere of radius r can be given by spherical coordinates (r, θ, ϕ) , $\theta \in [0, 2\pi]$ and $\phi \in$ _____. (1)
- 19) If $\delta(t - a)$ is the Dirac-delta function, then $\int_{-\infty}^{\infty} e^{7t} \delta(t - 5) dt =$ _____. (1)

SECTION-I

Answer all the questions.

Q-II Answer the following questions as directed.

- 21) Find the derivative of function $\frac{x^2 e^x - \sin x}{\log x}$. (2)
- 22) What is the greatest rate of increase of the function $u = xyz^2$ at the point (1,0,3)? (2)

a)

- [OR]
b) If $\vec{x} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{y} = 3\hat{i} - \hat{j} + 2\hat{k}$, show that $\vec{x} + \vec{y}$ is perpendicular to $\vec{x} - \vec{y}$. (2)
- 23) Write the spherical and cylindrical coordinates. (2)
- 24) Find the Taylor's series expansion of function $\cos x$ about $x = 0$. (3)
- 25) Define: Dirac delta function $\delta(t)$. Show that $\int_{-\infty}^{\infty} f(t)\delta'(t-a)dt = -f'(a)$. (2)
- 26) A microchip company has two machines that produce the chips. Machine I produces 65% of the chips, but 5% of its are defective. Machine II produces 35% of the chips and 15% of its chips are defective. A chip is selected at random and found to be defective. What is the probability that it came from machine I? (2)
- 27) In sampling a large number of parts manufactured by a machine, the mean number of defectives in a sample of 20 is 2. Out of 1000 such samples, how many would be expected to contain at least 3 defective parts? (4)
- a)
- [OR]
b) The mean height of 500 students is 150 cm and the standard deviation is 15 cm. Assuming that the heights are normally distributed, find how many students's heights lie between 120 cm and 155 cm. (4)
- 28) Find the work done when a force $\vec{F} = (x^2 - y^2 + x)\hat{i} - (2xy + y)\hat{j}$ moves a particle from origin to (1,1) along a parabola $y^2 = x$. (3)
- a)
- [OR]
b) Find a unit normal vector to the surface $x^2 + y^2 + z^2 = 5$ at the point (0,1,2). (3)

SECTION-II

Answer all the questions.

Q-III Answer the following questions as directed.

- 29) The pressure P at any point (x, y, z) in space is $P = 400xyz^2$. Find the highest pressure at the surface of a unit sphere $x^2 + y^2 + z^2 = 1$. (5)
- 30) Using Stoke's theorem, evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = y^2\hat{i} + x^2\hat{j} - (x+z)\hat{k}$ and C is the boundary of the triangle with vertices at (0,0,0), (1,0,0) and (1,1,0). (5)
- a)
- [OR]
b) Find the directional derivative of the scalar function $f(x, y, z) = xyz$ in the direction of the outer normal to the surface $z = xy$ at the point (3,1,3). (5)
- 31) Find $\iint_S \vec{F} \cdot \hat{n} ds$, where $\vec{F} = (2x + 3z)\hat{i} - (xz + y)\hat{j} + (y^2 + 2z)\hat{k}$ and S is the surface of the sphere having center (3,1,2) and radius 3. (5)
- a)
- [OR]
b) State Green's theorem for a plane. Using it, evaluate the integral $\oint_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$, where C is the boundary of the region defined by $y = \sqrt{x}$ and $y = x^2$. (5)
- 32) A fluid motion is given by $\vec{v} = (y+z)\hat{i} + (z+x)\hat{j} + (x+y)\hat{k}$. Show that the motion is irrotational and hence find the velocity potential. (5)
- 33) Solve the differential equation $\frac{dy}{dx} + xy = x^3y^3$. (5)

a) [OR] Using method of variation of parameteres, solve the differential equation (5)

b) $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 8y = e^{-5x}.$

34) What is the necessary and suffcicient condition for the differential equation (5)
 $M(x,y)dx + N(x,y)dy = 0$ to be exact? Show that a differential equation

a) $(5x^4 + 3x^2y^2 - 2xy^3)dx + (2x^3y - 3x^2y^2 - 5y^4)dy = 0.$ Also solve it.

[OR] An inductance of 2 henries and a resistance of 20 ohms are connected in series with an e.m.f. E (5)
b) volts. If the current is zero when $t = 0$, find the current at the end of 0.01 sec if $E = 100$ volts.

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